# Boréal Marché — Customer Intelligence Platform

*Plateforme d'intelligence client · une chaîne de données, trois modèles, un service gouverné*

One data pipeline, three model heads — **retention**, **basket recommendations**, a **bilingual assistant** — deployed, monitored and governed under Québec's Law 25.

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Live: https://boreal-marche-capstone-4ay3m2fn6qdo4jtgy72vpr.streamlit.app · API: https://padp32qm6p.us-east-1.awsapprunner.com/docs
Recorded talk: https://youtu.be/2bcNpRwMQxE

# Demonstration first

*Démonstration d'abord — le système en service, commit 422bbab*

1. **Retention** — a customer → P(order in 90 days), the *proposed* coupon, and the three factors behind it
2. **Basket** — herb markers → rules fire with their lift; an unknown basket → honest "best-sellers" fallback
3. **Assistant** — a question in French → answer with its cited policy; a request for another customer's data → refused
4. **Monitoring** — the shifted cohort turns drift red; alerts fired

| Check before the talk | |
|---|---|
| `/version` = deployed commit | `422bbab` |
| `/ready` | 200 |
| Tests in CI | 89, coverage 82% |
| Fallback video | https://youtu.be/C_Qv7lu3cpY |

# The problem: a coupon that loses money

*Le problème : un coupon envoyé à tous qui fait perdre de l'argent*

Every quarter, a 15% coupon goes to **every** active customer.

**−$55,176**

result of that policy on the last held-out quarter — mostly discounts given to loyal customers who would have ordered anyway.

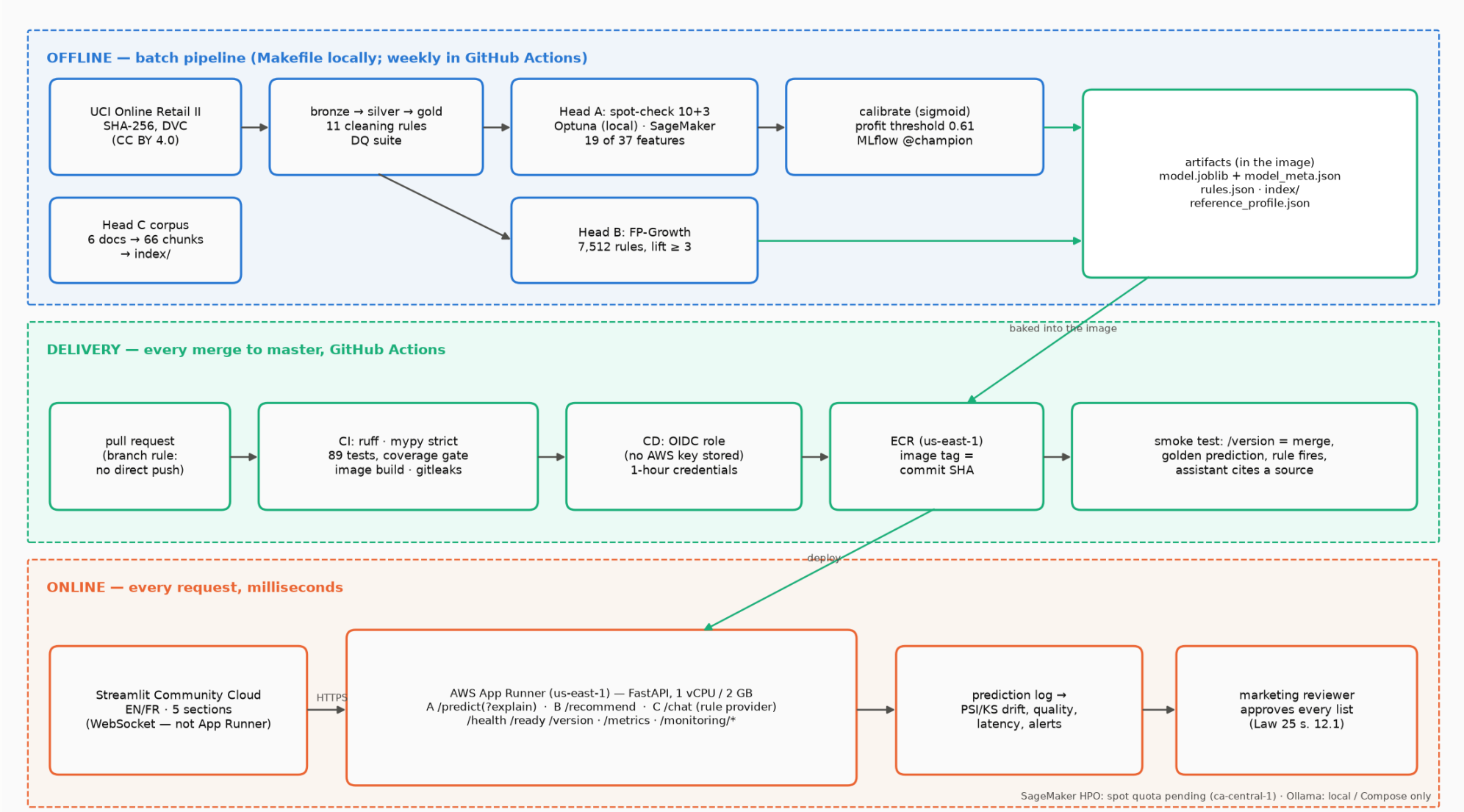
Three client questions, one system:

- **A — Retention:** who is about to stop ordering, and is a coupon worth it?
- **B — Cross-sell:** what else belongs in this basket?
- **C — Support:** answer the 40 recurring questions in FR/EN without inventing anything

Data: UCI Online Retail II, 1,067,371 lines, CC BY 4.0, re-skinned as Boréal Marché.

# Architecture: artifacts cross the line, algorithms do not

*Architecture : seuls les artefacts passent du hors-ligne au service en ligne*



# Trustworthy data before any model

*Des données fiables avant tout modèle*

- **11 cleaning rules, counted:** 64,033 lines removed (6.0%); the largest is a **22,523-line overlap** between the two yearly sheets that the starter script double-counted
- **Point-in-time features:** for cut-off T, features see only orders before T; label = an order in [T, T + 90 d)
- **Leakage tested on purpose:** features computed through the label window → PR-AUC 0.8070 → **1.0000**. A perfect score is a bug — unit tests now fence it off
- **Test set protected:** scored 3 times, each after every choice was fixed, every look disclosed

# Head A results: far above the business rule

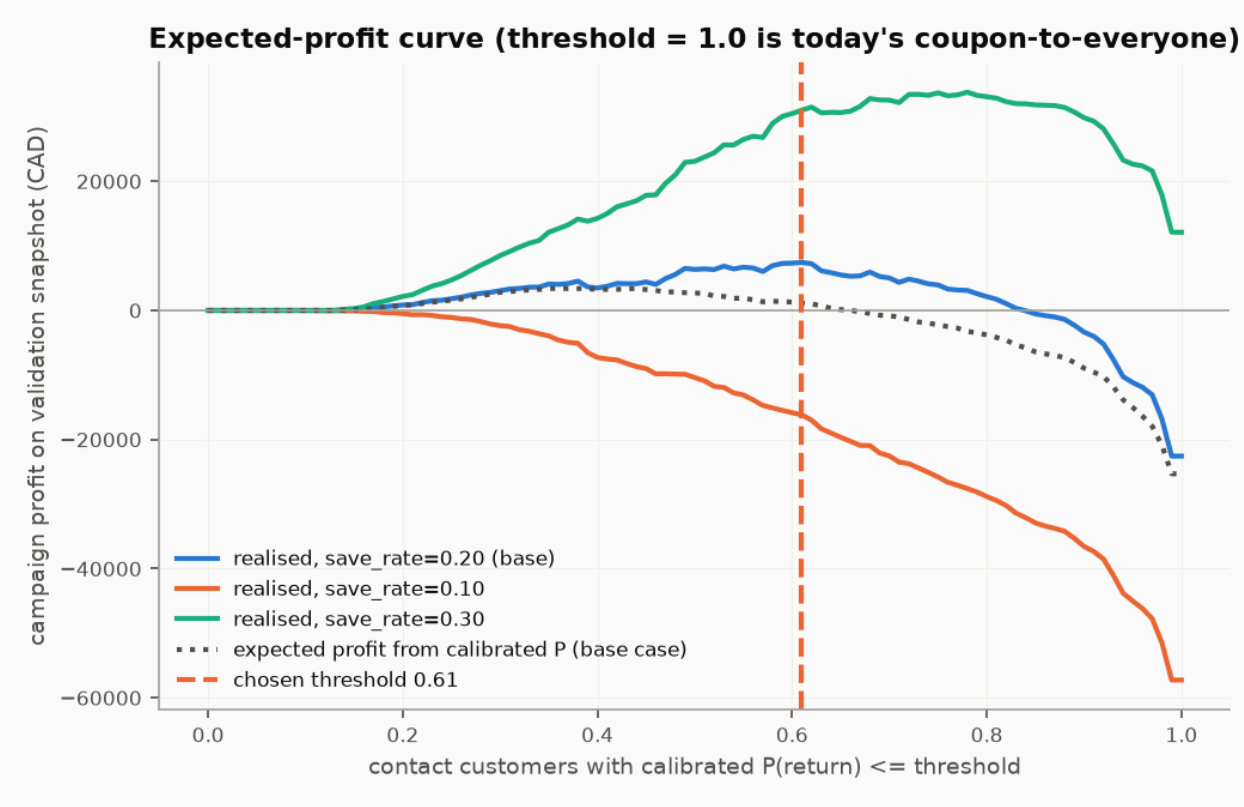
*Rétention : nettement au-dessus de la règle d'affaires*

| | PR-AUC | 95% CI (bootstrap) | ROC-AUC | Brier | Recall@top10% |
|---|---|---|---|---|---|
| Model (XGBoost, calibrated) | 0.8472 | 0.832–0.863 | 0.7675 | 0.1916 | 0.161 |
| Baseline: Dummy (prior) | 0.6069 | 0.586–0.625 | 0.5000 | 0.2565 | 0.126 |
| Baseline: rule 'ordered in last 60 days' | 0.6734 | 0.649–0.695 | 0.6187 | 0.2439 | 0.151 |
| Heuristic: recency deciles | 0.7123 | 0.687–0.734 | 0.6563 | 0.2416 | 0.132 |

- Test quarter 2011-09-01, 2,768 customers, **19 features** (of 37: the other 18 were unnecessary and are no longer stored)
- Calibrated: ECE 0.041 on test — the probability can be multiplied by dollars
- Ten model families were within noise of each other; tuning added +0.0048. **The limit is the features, not the algorithm.**

# Hard decision (1/3): a probability is not a decision

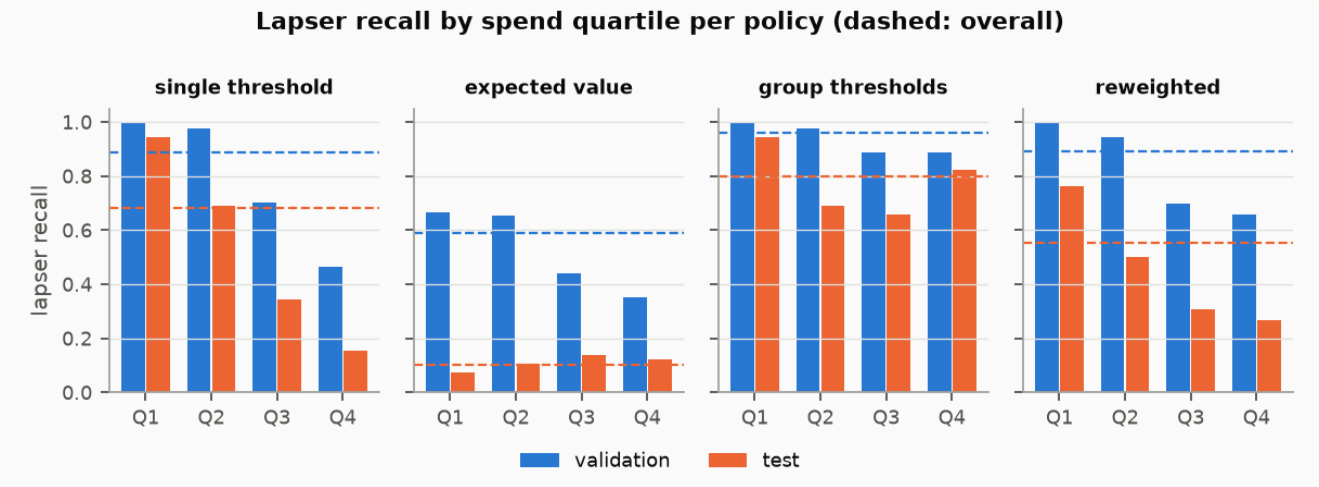
*Décision difficile : une probabilité n'est pas une décision*




- Coupon if P(return) ≤ **0.61**, the profit-maximising point on validation
- Lapser: `save × (value − discount) − coupon`; loyal: `−(coupon + discount)`
- **First design failed:** valuing a save at 90 days of margin made "contact nobody" optimal — the assumption was wrong, not the model
- At a 10% save rate **no** campaign pays

# Hard decision (2/3): one threshold misses the best customers

*Un seul seuil oublie nos meilleurs clients*




Lapser recall: **89%** overall, **46%** in the top spend quartile.

58 high spenders who did leave were never proposed — **$74,672** of margin.

The model ranks them well; the **single value-blind threshold** is the cause.

# Hard decision (3/3): what I chose, and why

*Ce que j'ai choisi, et pourquoi*

| Option | Fixes | Costs |
|---|---|---|
| M1 expected value | most profitable | abandons most lapsers; depends on the unmeasured save rate |
| M2 threshold per spend group | closes the gap | 20% more coupons, mostly to customers who stay; treatment set by spend |
| M3 re-weighting | halves the gap | the gain shrinks on test |

**Chosen:** keep one threshold for the automatic proposal **+** a *high-value review list* for the manager who already approves every list (Law 25 s. 12.1: a person reviews) **+** a randomised 10% hold-out to measure the save rate per group. The real unknown is the coupon's effect, not the model.

# Heads B and C

*Recommandations et assistant bilingue*

**B — association rules** (7,512 rules)

- Hit-rate@5 **0.145** vs 0.017 for best-sellers: **8.6×**
- 91.7% of baskets served by a rule; every suggestion names its rule
- Lift trap: confidence 44.7%, lift 2.998 → removed

**C — retrieval + typed tools**

- Tool choice: local LLM **97.5%** golden, **95%** paraphrases
- Rule baseline 100% → **35%** on paraphrases: its keywords overfit
- Out-of-scope refused 100%; never invents a SKU or a probability

# Engineering: deployed, monitored, cheap

*Ingénierie : déployé, surveillé, peu coûteux*

- CI: ruff, mypy strict, 89 tests, image build, gitleaks
- CD: **OIDC** — no AWS key in GitHub; image tag = commit; smoke test checks `/version`
- Image 823 MB (was 1,437 MB); `/predict` p50 4.8 ms
- Retraining opens a **pull request**, never deploys

- Shifted cohort: 53% of features red, 18 alerts
- Total AWS spend: **$3.05**
- SageMaker HPO: spot quota pending (case 179073448000753, opened 2026-09-29) — tuning ran locally with Optuna

# Limitations — stated, not hidden

*Limites — déclarées, pas cachées*

- **Propensity, not uplift:** every dollar rests on a 20% save rate never measured
- **One test quarter**, public re-skinned data; regions are a synthetic mapping
- **One-order customers** are close to coin flips (PR-AUC 0.594)
- **Fairness gap open** — the review list and hold-out are designed, not built
- Live assistant runs the rule provider; hosted LLM never evaluated
- Serving in `us-east-1` (App Runner is not offered in Canada); SQLite log lost on redeploy

# What I would do next

*Prochaines étapes*

1. **Measure the coupon:** 10% randomised hold-out for two quarters → uplift model
2. **Build the high-value review list** and audit subgroups on backfilled labels
3. **Canada region** (ECS Fargate, `ca-central-1`) and Postgres before any real customer data
4. **Run SageMaker HPO** when the quota is granted; compare cost and score with local Optuna
5. **Onboarding rule** for one-order customers instead of a model that cannot rank them

> Recommendation to the client: stop the blanket coupon; pilot the reviewed, targeted list — $120k–$217k a year under stated assumptions.

# Merci · Thank you — Questions?

*Interface : https://boreal-marche-capstone-4ay3m2fn6qdo4jtgy72vpr.streamlit.app*

*API : https://padp32qm6p.us-east-1.awsapprunner.com/docs*

*Repository: github.com/OlgaMorgunova/boreal-marche-capstone*

*Recorded talk / présentation enregistrée : https://youtu.be/2bcNpRwMQxE*